FINAL Unit2 MCQ - Topic order

1. Users of a mobile application are categorized based on the average number of hours they use the application each day, as shown in the following table.

Category	Average Hours of Use Per Day
Top User	Greater than 5
Regular User	Between 1 and 5, inclusive
Inactive User	Less than 1

Suppose *n* is the average number of hours a user spends using the application each day. Which of the following algorithms can be used to assign the correct categorization to a user?

(A) If *n* is greater than 5, categorize the user as a top user. Otherwise, if *n* is less than or equal to 5, categorize the user as a regular user. Otherwise, categorize the user as an inactive user.

(B) If *n* is greater than 5, categorize the user as a top user. Otherwise, if *n* is greater than or equal to 1, categorize the user as a regular user. Otherwise, categorize the user as an inactive user.

(C) If *n* is less than 1, categorize the user as an inactive user. Otherwise, if *n* is greater than or equal to 1, categorize the user as a regular user. Otherwise, categorize the user as a top user.

(D) If *n* is greater than or equal to 1, categorize the user as a regular user. Otherwise, if *n* is less than 1, categorize the user as an inactive user. Otherwise, categorize the user as a top user.

2. Which of the following algorithms can be used to display exactly 10 random even numbers between 0 and 100, inclusive?

(A) Repeat the following 100 times: generate a random number between 0 and 10, inclusive. If the number is even, display it.

(B) Repeat the following 10 times: generate a random number between 0 and 100, inclusive, multiply it by 2, and display the result.

(C) Repeat the following 10 times: generate a random number between 0 and 100, inclusive. If the number is even, display it.

(D) Repeat the following 10 times: generate a random number between 0 and 50, multiply it by 2, and display the result.

3. A program is being written to control access to a secure facility. The gate to the facility should open only when the user has a valid access badge and the current hour is between 7 and 18, inclusive.

Which of the following algorithms can be used to determine whether to allow access to the facility?

(A) If the person has a valid access badge and the hour is at least 7 and at most 18, then open the gate; otherwise, deny access.

(B) If the person has a valid access badge and the hour is greater than 7 and less than 18, then open the gate; otherwise, deny access.

(C) If the person does not have a valid access badge and the hour is at least 7 and at most 18, then deny access; otherwise, open the gate.

(D) If the person does not have a valid access badge and the hour is less than 7 and greater than 18, then deny access; otherwise, open the gate.

4. A company is designing a program to evaluate the strength of a user's password. The program needs to indicate if the password is "weak," "medium," or "strong" based on the following guidelines.

- A password with fewer than 8 characters is classified as "weak."
- A password with 8 or more characters is classified as "medium," where all characters are either numbers or letters.
- A password with 8 or more characters plus at least one special character (like !, @, or #) is classified as "strong."

Which of the following algorithms will correctly classify a password?

(A) Step 1: If the password has fewer than 8 characters, label it as "weak" and skip Steps 2 and 3. Otherwise, go to Step 2.

Step 2: If the password has more than 8 characters, label it as "medium" and skip Step 3. Otherwise, go to Step 3.

Step 3: Label the password as "strong."

(B) Step 1: If the password has fewer than 8 characters, label it as "weak" and skip Steps 2 and 3. Otherwise, go to Step 2.

Step 2: If the password contains at least one special character, label it as "strong" and skip Step 3. Otherwise, go to Step 3.

Step 3: Label the password as "medium."

(C) Step 1: If the password has 8 or more characters, label it as "medium" and skip Steps 2 and 3. Otherwise, go to Step 2.

Step 2: If the password has more than 8 characters and contains at least one special character, label it as "strong" and skip Step 3. Otherwise, go to Step 3.

Step 3: Label the password as "weak."

(D) Step 1: If the password has 8 or more characters and contains at least one special character, label it as "strong" and skip Steps 2 and 3. Otherwise, go to Step 2.

Step 2: If the password does not contain at least one special character, label it as "medium" and skip Step 3. Otherwise, go to Step 3.

Step 3: Label the password as "weak."

5. A programmer wants to create a fitness application that counts the number of days in a month that a user walks at least 10,000 steps. Which of the following algorithms can be used to determine the correct count?

(A) At the beginning of the month, start with a day count of 0 and a total step count of 0. At the end of each day, add 1 to the day count, check the number of steps taken that day, and add it to the total step count. At the end of the month, if the total step count is at least 10,000, output the day count.

(B) At the beginning of the month, start with a day count of 0. At the end of each day, check the number of steps taken. Set the day count equal to the number of steps taken that day. At the end of the month, if the day count is at least 10,000, output the day count.

(C) At the beginning of the month, start with a day count of 0. At the end of each day, check the number of steps taken. If the number of steps taken that day is at least 10,000, set the day count equal to the number of steps taken that day. At the end of the month, output the day count.

(D) At the beginning of the month, start with a day count of 0. At the end of each day, check the number of steps taken. If the number of steps taken that day is at least 10,000, add 1 to the day count. At the end of the month, output the day count.

6. The variables x, y, and z are integers. Which of the following algorithms can be used to swap the values of the variables so that they are ordered from least to greatest (i.e., $x \le y \le z$)?

(A)

Step 1:	If x is greater than y, swap the values of x and y.
Step 2:	If y is greater than z, swap the values of y and z.

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(B)

Step 1:	If x is greater than z, swap the values of x and z.
Step 2:	If y is greater than z, swap the values of y and z.

(C)

Step 1:	If x is greater than y, swap the values of x and y.
Step 2:	If y is greater than z, swap the values of y and z.
Step 3:	If x is greater than y, swap the values of x and y.

(D)

Step 1:	If x is greater than y, swap the values of x and y.
Step 2:	If y is greater than z, swap the values of y and z.
Step 3:	If x is greater than z, swap the values of x and z.

7. Consider the following declarations.

```
int valueOne, valueTwo;
```

Assume that `valueOne` and `valueTwo` have been initialized. Which of the following evaluates to `true` if `valueOne` and `valueTwo` contain the same value?

(A) `valueOne.equals((Object) valueTwo)`
(B) `valueOne == valueTwo`
(C) `valueOne.compareTo(valueTwo) == 0`
(D) `valueOne.equals(valueTwo)`

8. In the following code segment, a and b are properly declared and initialized boolean variables.

```
boolean result = (a != b);
System.out.println(result);
```

Which of the following best describes the behavior of the code segment?

(A) It prints true if a is true and b is false, and prints false otherwise.
(B) It prints true if a and b have different values, and prints false otherwise.
(C) It prints true if a and b have the same value, and prints false otherwise.
(D) It prints false for all possible values of a and b.

9. Consider the following code segment.

```
int a = 10;
int b = 5 * 2;
```

```
System.out.print(a == b);
```

What is printed as a result of executing the code segment?

(A)
10

(B)
10 == 10

(C)
true

(D)
false

10. Consider the following code segment, which is intended to print `true` only when both `r` and `v` are positive numbers. Assume that `r` and `v` have been properly declared and initialized.

```
boolean rPos = r > 0;
boolean vPos = v > 0;
System.out.print(rPos == vPos);
```

The code segment does not always work as intended.

Which of the following values for `r` and `v` will demonstrate that this code segment does **not** work as intended?

(A)
r = -10, v = 10

(B)
r = -1, v = -4

(C)
r = 5, v = -2

(D)
r = 6, v = 3

11. Consider the following statement.

```
boolean x = (5 < 8) == (5 == 8);
```

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What is the value of x after the statement is executed?

(A)
5

(B)
8

(C)
true

(D)
false

12. In the following Boolean expression, the int variables x and y have been properly declared and initialized.

$(x <= 10) == (y > 25)$

Which of the following values for x and y will result in the expression evaluating to true?

(A) x = 8 and y = 25
(B) x = 10 and y = 10
(C) x = 10 and y = 30
(D) x = 15 and y = 30

13. Consider the following incomplete method, which is intended to return `true` if the value of `y` is between the values of the other two parameters and `false` otherwise.

```
/** Precondition: x, y, and z have 3 different values. */
public static boolean compareThree(int x, int y, int z)
{
 return /* missing condition */ ;
}
```

The following table shows the results of several calls to `compareThree`.

Call	Result
`compareThree(4, 5, 6)`	`true`
`compareThree(6, 5, 4)`	`true`
`compareThree(5, 4, 6)`	`false`
`compareThree(3, 4, 4)`	violates precondition

Which of the following can be used to replace `/* missing condition */` so that `compareThree` will work as intended when called with parameters that satisfy its precondition?

(A) `(x > y) && (y > z)`

(B) `(x > y) || (y > z)`

(C) `(x > y) == (y > z)`

(D) `(x > y) != (y > z)`

14. In the following statement, j and k are properly declared and initialized int variables.

boolean result = ((j > 0) && (k > 0)) || ((j < 0) && (k < 0));

Which of the following best describes the conditions in which result is assigned the value true?

(A) When j and k are both equal to 0
(B) When j and k are both positive or both negative
(C) When j is greater than k
(D) When j is positive and k is negative or when j is negative and k is positive

15. Consider the following code segment.

```
int x = /* some integer value */ ;
int y = /* some integer value */ ;

boolean result = (x < y);

result = ( (x >= y) && !result );
```

Which of the following best describes the conditions under which the value of `result` will be `true` after the code segment is executed?

(A) Only when `x >= y`
(B) Only when `x` and `y` are equal
(C) The value will always be `true`.
(D) The value will never be `true`.

16. Consider the following method.

```
public int someCode(int a, int b, int c)
{
  if ((a < b) && (b < c))
    return a;
  if ((a >= b) && (b >= c))
    return b;
  if ((a == b) || (a == c) || (b == c))
```

```
        return c;
    }
```

Which of the following best describes why this method does not compile?

(A) It is possible to reach the end of the method without returning a value.
(B) The `if` statements must have `else` parts when they contain `return` statements.
(C) Methods cannot have multiple `return` statements.
(D) The third `if` statement is not reachable.

17. Consider the following code segment, in which `m` and `n` are properly declared and initialized `int` variables.

```
int result = 0;
for (int i = 0; i < n; i++)
{
  for (int j = m; j < n; j++)
  {
    result++;
  }
}
```

Which of the following best describes the conditions that must be true of m and n so that after this code segment executes, the value of result is greater than 0?

(A) $n > 0$ and $m > 0$
(B) $n > 0$ and $m > n$
(C) $n > 0$ and $n > m$
(D) $m > 0$ and $m > n$

18. The following code segment is intended to print the total cost of a stay in a room at a particular hotel. The int variable numNights represents the length of the stay, in nights, and the double variable dailyRate represents the room base cost for each night of the stay. For stays that are longer than five nights, there is a 10% discount applied to the room base cost for all nights during the stay. In addition to the room cost, the hotel charges a $25 resort fee for each night of the stay. The resort fee is not eligible for the 10% discount for stays that are longer than five nights.

```
int numNights =   /* some initial value */ ;
double dailyRate = /* some initial value */ ;

double totalCost = numNights * dailyRate;
/* missing code */

System.out.println(totalCost);
```

Which of the following can be used to replace /* missing code */ so that the code segment works as intended?

(A)

```
if (numNights > 5)
{
  totalCost *= 0.9;
  totalCost += (numNights * 25);
}
```

(B)

```
if (numNights > 5)
{
  totalCost *= 0.9;
}
totalCost += (numNights * 25);
```

(C)

```
totalCost += (numNights * 25);
if (numNights > 5)
{
  totalCost *= 0.9;
}
```

(D)

```
totalCost += 25;
if (numNights > 5)
{
  totalCost *= 0.9;
}
```

19. Consider the following code segment, which is intended to simulate a random process. The code is intended to set the value of the variable `event` to exactly one of the values `1`, `2`, or `3`, depending on the probability of an event occurring. The value of `event` should be set to `1` if the probability is 70 percent or less. The value of `event` should be set to `2` if the probability is greater than 70 percent but no more than 80 percent. The value of `event` should be set to `3` if the probability is greater than 80 percent. The variable `randomNumber` is used to simulate the probability of the event occurring.

```
int event = 0;
if (randomNumber <= 0.70)
{
event = 1;
}
```

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```
if (randomNumber <= 0.80)
{
event = 2;
}
else
{
event = 3;
}
```

The code does not work as intended. Assume that the variable `randomNumber` has been properly declared and initialized. Which of the following initializations for `randomNumber` will demonstrate that the code segment will not work as intended?

(A) `randomNumber = 0.70;`

(B) `randomNumber = 0.80;`

(C) `randomNumber = 0.85;`

(D) `randomNumber = 1.00;`

20. Consider the following code segment.

```
int temp = 75;
String choice;
boolean hasSibling = false;

if (temp < 70)
{
  choice = "Indoors";
}
else
{
  choice = "Outdoors";
}
if (hasSibling)
{
  choice += " with sibling";
}
```

What is the value of `choice` after executing the code segment?

(A)
"Indoors"

(B)
"Outdoors"

(C)
"Indoors with sibling"

(D)
"Outdoors with sibling"

21. Consider the following code segment.

```
int x = 3;
int y = -1;

if (x - 2 > y)
{
  x -= y;
}
if (y + 3 >= x)
{
  y += x;
}

System.out.print("x = " + x + ", y = " + y);
```

What is printed as a result of executing the code segment?

(A)
x = 2, y = 1

(B)
x = 3, y = 2

(C)
x = 4, y = -1

(D)
x = 4, y = 3

22. Consider the following two code segments where the int variable choice has been properly declared and initialized.

Code Segment 1

```
if (choice > 10)
{
  System.out.println("blue");
}
else if (choice < 5)
{
```

```
  System.out.println("red");
}
else
{
  System.out.println("yellow");
}
```

Code Segment 2

```
if (choice > 10)
{
  System.out.println("blue");
}
if (choice < 5)
{
  System.out.println("red");
}
else
{
  System.out.println("yellow");
}
```

Assume that both code segments initialize choice to the same integer value. Which of the following best describes the conditions on the initial value of the variable choice that will cause the two code segments to produce different output?

(A) choice > 5 and choice < 10
(B) choice > 10
(C) choice == 5 or choice == 10
(D) There is no value for choice that will cause the two code segments to produce different output.

23. Consider the following method.

```
public String exercise(int input)
{
if (input < 10)
{
    return "alpha";
}
if (input < 5)
{
    return "beta";
}
if (input < 1)
{
    return "gamma";
}
return "delta";
```

```
}
```

Assume that the `int` variable `x` has been initialized in another method in the same class. Which of the following describes the conditions under which the method call `exercise(x)` will return `"gamma"` ?

(A) The method will never return `"gamma"`.

(B) The method will return `"gamma"` if `x` is less than `1`.

(C) The method will return `"gamma"` if `x` is between `1` and `4`, inclusive.

(D) The method will always return `"gamma"`.

24. The following method is intended to return `true` if and only if the parameter `val` is a multiple of `4` but is not a multiple of `100` unless it is also a multiple of `400`. The method does not always work correctly.

```
public boolean isLeapYear(int val)
{
if ((val % 4) == 0)
{
    return true;
}
else
{
    return (val % 400) == 0;
}
}
```

Which of the following method calls will return an incorrect response?

(A) `isLeapYear(1900)`

(B) `isLeapYear(1984)`

(C) `isLeapYear(2000)`

(D) `isLeapYear(2001)`

25. In the code segment below, the `int` variable `temp` represents a temperature in degrees Fahrenheit. The code segment is intended to print a string based on the value of `temp`. The following table shows the string that should be printed for different temperature ranges.

Temperature Range	String to Print
31 and below	`"cold"`
32–50	`"cool"`
51–70	`"moderate"`
71 and above	`"warm"`

```
String weather;
if (temp <= 31)
{
```

```
weather = "cold";
}
else
{
weather = "cool";
}
if (temp >= 51)
{
weather = "moderate";
}
else
{
weather = "warm";
}
System.out.print(weather);
```

Which of the following test cases can be used to show that the code does NOT work as intended?

I. `temp = 30`
II. `temp = 51`
III. `temp = 60`

(A) I only
(B) I and II only
(C) II and III only
(D) I, II, and III

26. A school that does not have air conditioning has published a policy to close school when the outside temperature reaches or exceeds 95°F. The following code segment is intended to print a message indicating whether or not the school is open, based on the temperature. Assume that the variable `degrees` has been properly declared and initialized with the outside temperature.

```
if (degrees > 95)
{
System.out.println("School will be closed due to extreme heat");
}
else
{
System.out.println("School is open");
}
```

Which of the following initializations for `degrees`, if any, will demonstrate that the code segment may not work as intended?

(A) `degrees = 94;`
(B) `degrees = 95;`
(C) `degrees = 96;`
(D) The code will work as intended for all values of `degrees`.

27. Consider the following code segment.

```
int x = 7;
if (x < 7)
{
 x = 2 * x;
}
if (x % 3 == 1)
{
 x = x + 2;
}
System.out.print(3 * x);
```

What is printed as a result of executing the code segment?

(A) 9
(B) 14
(C) 21
(D) 27

28. In the following code segment, n is a properly declared and initialized int variable.

```
boolean result = false;
if (n >= 10)
{
  result = true;
}
if (n <= 50)
{
  result = true;
}
System.out.println(result);
```

Which of the following best describes the behavior of the code segment?

(A) It prints true for all possible values of n.
(B) It prints false for all possible values of n.
(C) It prints true if n is between 10 and 50, inclusive, and prints false otherwise.
(D) It prints false if n is between 10 and 50, inclusive, and prints true otherwise.

29. The following method is intended to return "Renaissance" for years from 1400 to 1600, inclusive, "Medieval" for years from 400 to 1399, inclusive, or "Other" for any other year. In the method, the parameter represents a year.

```
public static String getCategory(int year)
{
  String category = "";
```

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```
  /* missing code */
  return category;
}
```

Which of the following code segments can replace /* missing code */ so that the getCategory method works as intended?

(A)

```
if (year > 1600 || year < 400)
{
  category = "Other";
}
else if (year >= 1400)
{
  category = "Renaissance";
}
else if (year >= 400)
{
  category = "Medieval";
}
```

(B)

```
if (year > 1600 || year < 400)
{
  category = "Other";
}
if (year >= 1400)
{
  category = "Renaissance";
}
if (year >= 400)
{
  category = "Medieval";
}
```

(C)

```
if (year >= 1400)
{
  category = "Renaissance";
}
else if (year >= 400)
{
  category = "Medieval";
}
else
{
  category = "Other";
}
```

(D)

```
if (year >= 1400)
{
  category = "Renaissance";
}
if (year >= 400)
{
  category = "Medieval";
}
else
{
  category = "Other";
}
```

30. A computer game should print different messages based on a player's score.

- "Great!" is printed for scores greater than 500.
- "Good!" is printed for scores between 250 and 500, inclusive.
- "Try again." is printed for scores less than 250.

Assume that score is a properly declared and initialized int that represents a player's score.

Which of the following code segments will print the correct message for any valid value of score?

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(A)

```
String message = "";
if (score < 250)
{
  message = "Try again.";
}
else if (score <= 500)
{
  message = "Good!";
}
else
{
  message = "Great!";
}
System.out.println(message);
```

(B)

```
String message = "";
if (score < 250)
{
  message = "Try again.";
}

if (score <= 500)
{
  message = "Good!";
}
else
{
  message = "Great!";
}
System.out.println(message);
```

(C)

```
String message = "";
if (score >= 250)
{
  message = "Good!";
}
else if (score > 500)
{
  message = "Great!";
}
else
{
  message = "Try again.";
}
System.out.println(message);
```

(D)

```
String message = "";
if (score > 500)
{
  message = "Great!";
}
else if (score <= 500)
{
  message = "Good!";
}
else
{
  message = "Try again.";
}
System.out.println(message);
```

31. Consider the following two code segments. Assume that variables x and y have been declared as int variables and have been assigned integer values.

Code Segment I

```
int result = 0;
if (x > y)
{
  result = x - y;
  System.out.print(result);
}
else if (x < y)
{
  result = y - x;
  System.out.print(result);
}
else
{
  System.out.print(result);
}
```

Code Segment II

```
if (x < y)
{
  System.out.print(y - x);
}
else
{
  System.out.print(x - y);
}
```

Which of the following correctly compares the outputs of the two code segments?

(A) Code segment I and code segment II produce the same output for all values of x and y.
(B) Code segment I and code segment II produce the same output only when x is equal to y.
(C) Code segment I and code segment II produce the same output only when x is not equal to y.
(D) Code segment I and code segment II produce the same output only when x is less than y.

32. Consider the following method, which returns an `int` based on its parameter `x`.

```
public static int puzzle(int x)
{
if (x > 20)
{
    x -= 2;
}
else if (x % 2 == 0) // Line 7
{
    x += 4;
}
return x;
```

```
}
```

Consider a modification to the method that eliminates the `else` from line 7 so that line 7 becomes

```
if (x % 2 == 0) // Modified line 7
```

For which of the following values of `x` would the return values of the original method and the modified method differ?

(A) `5`

(B) `14`

(C) `22`

(D) `25`

33. In the following code segment, the `int` variables `a`, `b`, `c`, and `d` have been properly declared and initialized.

```
if (a < b || c != d)
{
  System.out.println("dog");
}
else
{
  System.out.println("cat");
}
```

Which of the following code segments produces the same output as the given code segment for all values of `a`, `b`, `c`, and `d`?

(A)

```
if (a < b && c != d)
{
  System.out.println("dog");
}
else
{
  System.out.println("cat");
}
```

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(B)

```
if (a > b && c == d)
{
  System.out.println("cat");
}
else
{
  System.out.println("dog");
}
```

(C)

```
if (a >= b || c == d)
{
  System.out.println("cat");
}
else
{
  System.out.println("dog");
}
```

(D)

```
if (a >= b && c == d)
{
  System.out.println("cat");
}
else
{
  System.out.println("dog");
}
```

34. Consider the following code segment.

```
int k = 0;
while (k < 10)
{
  System.out.print((k % 3) + " ");
  if ((k % 3) == 0)
    k = k + 2;
```

```
    else
      k++;
  }
```

What is printed as a result of executing the code segment?

(A) `0 2 0 2 0 2`

(B) `0 2 1 0 2 1 0`

(C) `0 2 0 2 0 2 0`

(D) `0 1 2 1 2 1 2`

35. Consider the following code segments, which are each intended to convert grades from a 100-point scale to a 4.0-point scale and print the result. A grade of 90 or above should yield a 4.0, a grade of 80 to 89 should yield a 3.0, a grade of 70 to 79 should yield a 2.0, and any grade lower than 70 should yield a 0.0.

Assume that `grade` is an `int` variable that has been properly declared and initialized.

Code Segment 1

```
double points = 0.0;
if (grade > 89)
{
  points = 4.0;
}
else if (grade > 79)
{
  points = 3.0;
}
else if (grade > 69)
{
  points = 2.0;
}
else
{
  points = 0.0;
}
System.out.println(points);
```

Code Segment 2

```
double points = 0.0;
if (grade > 89)
{
  points = 4.0;
}
if (grade > 79)
{
  points = 3.0;
}
```

```
if (grade > 69)
{
  points = 2.0;
}
if (grade < 70)
{
  points = 0.0;
}
System.out.println(points);
```

Which of the following statements correctly compares the values printed by the two code segments?

(A) The two code segments print the same value only when `grade` is below `80`.
(B) The two code segments print the same value only when `grade` is `90` or above.
(C) Both code segments print the same value for all possible values of `grade`.
(D) The two code segments print different values for all possible values of `grade`.

36. The following categories are used by some researchers to categorize regions as urban, suburban, or rural based on population density.

- An urban region is a region with more than 3,000 people per square mile.
- A suburban region is a region with between 1,000 and 3,000 people, inclusive, per square mile.
- A rural region is a region with fewer than 1,000 people per square mile.

Consider the following method, which is intended to categorize a region as urban, suburban, or rural based on the population density of the region.

```
public static String getCategory(int density)
{
  /* missing code */
}
```

Which of the following code segments can replace /* missing code */ so the getCategory method works as intended?

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(A)

```
String category;
if (density > 3000)
{
  category = "urban";
}
else if (density >= 1000)
{
  category = "suburban";
}
else
{
  category = "rural";
}
return category;
```

(B)

```
String category;
if (density > 3000)
{
  category = "urban";
}
else if (density <= 3000)
{
  category = "suburban";
}
else
{
  category = "rural";
}
return category;
```

(C)

```
String category = "rural";
if (density > 3000)
{
  category = "urban";
}
if (density >= 1000)
{
  category = "suburban";
}
return category;
```

(D)

```
String category = "urban";
if (density < 1000)
{
  category = "rural";
}
if (density >= 1000)
{
  category = "suburban";
}
return category;
```

37. In the following code segment, str is a properly declared and initialized String variable.

```
int result = 0;
if (str.length() > 5)
{
  if (str.indexOf("A") < 0)
  {
    result = 1;
  }
  else if (str.indexOf("B") < 0)
  {
    result = 2;
  }
}
else if (str.indexOf("A") < 0)
{
```

```
  result = 3;
}
```

Which of the following code segments assigns the same value to result as the preceding code segment for all values of str?

(A)

```
int result = 0;
if (str.indexOf("A") < 0)
{
  result = 3;
}
else if (str.indexOf("B") < 0)
{
  result = 2;
}
else if (str.length() > 5);
{
  result = 1;
}
```

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(B)

```
int result = 0;
if (str.indexOf("A") < 0)
{
  if (str.length() > 5)
  {
    result = 1;
  }
  else if (str.indexOf("B") < 0)
  {
    result = 2;
  }
  else
  {
    result = 3;
  }
}
```

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(C)

```
int result = 0;
if (str.indexOf("A") < 0)
{
  if(str.length() > 5)
  {
    result = 1;
  }
  else
  {
    result = 3;
  }
}
else if (str.indexOf("B") < 0)
{
  if (str.length() > 5)
  {
    result = 2;
  }
}
```

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(D)

```
int result = 0;
if (str.indexOf("B") < 0)
{
  if (str.length() > 5)
  {
    result = 2;
  }
}
else if (str.indexOf("A") < 0)
{
  if (str.length() > 5)
  {
    result = 1;
  }
  else
  {
    result = 3;
  }
}
```

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38. Consider the following code segment.

```
double cost = 25.0;
double discount = 0.0;

if (cost >= 100.0)
{
  discount = 10.0;
}
else if (cost >= 50.0)
{
  discount = 5.0;
}
else
{
  discount = 2.0;
}

System.out.println(cost - discount);
```

What is printed as a result of executing this code segment?

(A)
25.0

(B)
23.0

(C)
20.0

(D)
15.0

39. Consider the following code segment.

```
int start = 4;
int end = 5;

if (start < end)
{
  if (end > 0)
  {
    start += 2;
    end++;
  }
  else
```

```
  {
    end += 3;
  }
}

if (start < end)
{
  if (end == 0)
  {
    end += 2;
    start++;
  }
  else
  {
    end += 4;
  }
}
```

What is the value of `end` after the code segment is executed?

(A)
5

(B)
6

(C)
9

(D)
10

40. Consider the following code segment.

```
int x = 7;
int y = 4;
boolean a = false;
boolean b = false;

if (x > y)
{
  if (x % y >= 3)
  {
    a = true;
    x -= y;
  }
  else
  {
```

```
    x += y;
  }
}

if (x < y)
{
  if (y % x >= 3)
  {
    b = true;
    x -= y;
  }
  else
  {
    x += y;
  }
}
```

What are the values of `a`, `b`, and `x` after the code segment is executed?

(A) `a = true, b = true, x = -1`
(B) `a = true, b = false, x = 3`
(C) `a = true, b = false, x = 7`
(D) `a = false, b = false, x = 11`

41. Consider the following code segment, which uses properly declared and initialized int variables x and y and the String variable result.

```
String result = "";
if (x < 5)
{
  if (y > 0)
  {
    result += "a";
  }
  else
  {
    result += "b";
  }
}
else if (x > 10)
{
  if (y < 0)
  {
    result += "c";
  }
  else if (y < 10)
  {
    result += "d";
  }
  result += "e";
```

```
}
result += "f";
```

What is the value of result after the code segment is executed if x has the value 15 and y has the value 5?

(A)
adf

(B)
d

(C)
def

(D)
ef

42. Vehicles are classified based on their total interior volume. The `classify` method is intended to return a vehicle classification `String` value based on total interior volume, in cubic feet, as shown in the table below.

Vehicle size class	Total interior volume
Minicompact	Less than 85 cubic feet
Subcompact	85 to 99 cubic feet
Compact	100 to 109 cubic feet
Mid-Size	110 to 119 cubic feet
Large	120 cubic feet or more

The `classify` method, which does not work as intended, is shown below.

```
public static String classify(int volume)
{
String carClass = "";
if (volume >= 120)
{
    carClass = "Large";
}
else if (volume < 120)
{
    carClass = "Mid-Size";
}
else if (volume < 110)
{
    carClass = "Compact";
}
else if (volume < 100)
```

```
    {
        carClass = "Subcompact";
    }
    else
    {
        carClass = "Minicompact";
    }
    return carClass;
    }
```

The `classify` method works as intended for some but not all values of the parameter `volume`. For which of the following values of `volume` would the correct value be returned when the `classify` method is executed?

(A) `80`
(B) `90`
(C) `105`
(D) `115`

43. Consider the following code segment.

```
if (false && true || false)
{
  if (false || true && false)
  {
    System.out.print("First");
  }
  else
  {
    System.out.print("Second");
  }
}

if (true || true && false)
{
  System.out.print("Third");
}
```

What is printed as a result of executing the code segment?

(A)
Second

(B)
Third

(C)
FirstThird

(D)
SecondThird

44. Consider the following method.

```
public void conditionalTest(int a, int b)
{
  if ((a > 0) && (b > 0))
  {
    if (a > b)
      System.out.println("A");
    else
      System.out.println("B");
  }
  else if ((b < 0) || (a < 0))
    System.out.println("C");
  else
    System.out.println("D");
}
```

What is printed as a result of the call `conditionalTest(3, -2)`?

(A) `A`
(B) `B`
(C) `C`
(D) `D`

45. In the following code segment, num is a properly declared and initialized int variable.

```
boolean b1 = true;
if (num > 0)
{
  if (num >= 100)
  {
    b1 = false;
  }
}
else
{
  if (num >= -100)
  {
    b1 = false;
  }
}
```

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Which of the following statements always assigns the same value to b2 as the code segment assigns to b1?

(A)

```
boolean b2 = (num > -100) && (num < 100);
```

(B)

```
boolean b2 = (num > -100) || (num < 100);
```

(C)

```
boolean b2 = (num < -100) && (num > 0 || num < 100);
```

(D)

```
boolean b2 = (num < -100) || (num > 0 && num < 100);
```

46. The following incomplete method is intended to return the minimum value among its three parameters.

```
public static int minOfThree(int a, int b, int c)
{
  /* missing code */
}
```

Which of the following can be used to replace /* missing code */ so that the method works as intended?

(A)

```
if (a <= b)
{
  if (a <= c)
  {
    return a;
  }
  else
  {
    return c;
  }
}
else
{
  return b;
}
```

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(B)

```
if (a <= b)
{
  if (a <= c)
  {
    return a;
  }
  else if (b <= c)
  {
    return b;
  }
}
else
{
  return c;
}
```

(C)

```
if (a <= b || a <= c)
{
  return a;
}
else if (b <= a || b <= c)
{
  return b;
}
else
{
  return c;
}
```

(D)

```
if (a <= b && a <= c)
{
  return a;
}
else if (b <= a && b <= c)
{
  return b;
}
else
{
  return c;
}
```

47. In the following code segment, num is a properly declared and initialized int variable.

```
if (num > 0)
{
  if (num % 2 == 0)
  {
    System.out.println("X");
  }
  else
  {
    System.out.println("Y");
  }
}
```

Which of the following best describes the behavior of the code segment?

(A) When num is a negative odd integer, "X" is printed; otherwise, "Y" is printed.
(B) When num is a positive even integer, "X" is printed; otherwise, "Y" is printed.
(C) When num is a positive even integer, "X" is printed; when num is a positive odd integer, "Y" is printed; otherwise, nothing is printed.
(D) When num is a positive odd integer, "X" is printed; when num is a positive even integer, "Y" is printed; otherwise, nothing is printed.

48. A fitness application categorizes daily step counts into different activity levels. The "Active" level is defined as any step count between 10,000 and 15,000, inclusive. The following code segment is intended to determine if a user's step count falls within the "Active" category.

```
if (steps > 10000 && steps < 15000)
{
```

```
  activityLevel = "Active";
}
```

Which of the following best explains the error, if any, in this code segment?

(A) There is no error in the code segment as it correctly identifies “Active” step counts.
(B) The condition should be if (steps >= 10000 && steps <= 15000) to properly include 10,000 and 15,000.
(C) The condition should be if (steps > 10000 || steps < 15000) to include the correct range of step counts.
(D) The condition should be if (steps >= 10000 || steps <= 15000) to include all step counts from 10,000 and above, as well as 15,000 and below.

49. A certain device requires a passcode to be used. The passcode must meet both of the following requirements.

- The passcode must be 4 digits long.
- The passcode must not start with a zero (0) digit.

Assume that pass is a properly declared and initialized int variable.

Which of the following expressions will evaluate to true if pass is an acceptable passcode?

(A)

```
pass >= 1000 || pass < 10000
```

(B)

```
pass > 1000 || pass <= 10000
```

(C)

```
pass >= 1000 && pass < 10000
```

(D)

```
pass > 1000 && pass < 10000
```

50. Consider the following code segment.

```
boolean a = true;
boolean b = true;
System.out.print((b || (!a || b)) + " ");
System.out.print(((!b || !a) && a) + " ");
System.out.println(!(a && b) && b);
```

What output is produced when this code segment is executed?

(A)

```
true true true
```

(B)
true false true

(C)
true false false

(D)
false true false

51. Consider the following `boolean` variables.

- `isRaining` is `true` if it is raining and is `false` otherwise.
- `isSunny` is `true` if it is sunny and is `false` otherwise.
- `isWarm` is `true` if it is warm and is `false` otherwise.

Assume all three variables have been properly declared and initialized.

Which of the following expressions evaluates to `true` if it is sunny or warm and it is not raining, and evaluates to `false` otherwise?

(A)
(isSunny || isWarm) && isRaining

(B)
(isSunny || isWarm) && !isRaining

(C)
!(isSunny || isWarm) || isRaining

(D)
!(isSunny || isWarm) || !isRaining

52. Which of the following best describes the value of the Boolean expression shown below?

```
a && !(b || a)
```

(A) The value is always `false`.
(B) The value is `true` when `a` has the value `false`, and is `false` otherwise.
(C) The value is `true` when `b` has the value `false`, and is `false` otherwise.
(D) The value is `true` when either `a` or `b` has the value `true`, and is `false` otherwise.

53. Assume that x, y, and z are boolean variables that have been properly declared and initialized.
Which of the following expressions is equivalent to !x || !y || z?

(A)
!(x && y) && z

(B)
!(x && y) || z

(C)
!(x || y) && z

(D)
!(x || y) || z

54. Assume that isEven, isPositive, and isPrime are boolean variables that have been properly declared and initialized. Which of the following expressions is equivalent to the expression !(isEven && isPositive) && isPrime?

(A)
(!isEven || !isPositive) && isPrime

(B)
(!isEven || !isPositive) && !isPrime

(C)
(!isEven && !isPositive) && isPrime

(D)
(!isEven && !isPositive) && !isPrime

55. Consider the following code segment.

```
int num1 = 0;
int num2 = 3;
while ((num2 != 0) && ((num1 / num2) >= 0))
{
  num1 = num1 + 2;
  num2 = num2 - 1;
}
```

What are the values of `num1` and `num2` after the while loop completes its execution?

(A) `num1 = 0, num2 = 3`
(B) `num1 = 8, num2 = -1`
(C) `num1 = 4, num2 = 1`
(D) `num1 = 6, num2 = 0`

56. Consider the following method.

```
public void numberCheck(int maxNum)
{
  int typeA = 0;
  int typeB = 0;
  int typeC = 0;

  for (int k = 1; k <= maxNum; k++)
  {
    if (k % 2 == 0 && k % 5 == 0)
     typeA++;
    if (k % 2 == 0)
      typeB++;
    if (k % 5 == 0)
      typeC++;
  }

  System.out.println(typeA + " " + typeB + " " + typeC);
}
```

What is printed as a result of the call `numberCheck(50)` ?

(A) `5 20 5`
(B) `5 20 10`
(C) `5 25 5`
(D) `5 25 10`

57. Consider the following code segment.

```
String str = "a black cat sat on a table";
int counter = 0;

for (int i = 0; i < str.length() - 1; i++)
{
  if (str.substring(i, i + 1).equals("a") &&
     !str.substring(i + 1, i + 2).equals("b"))
  {
    counter++;
  }
}

System.out.println(counter);
```

What is printed as a result of executing the code segment?

(A)
1

(B)
2

(C)
3

(D)
5

58. Consider the following code segment.

```
String str1 = new String("Advanced Placement");
String str2 = new String("Advanced Placement");
if (str1.equals(str2) && str1 == str2)
{
System.out.println("A");
}
else if (str1.equals(str2) && str1 != str2)
{
System.out.println("B");
}
else if (!str1.equals(str2) && str1 == str2)
{
System.out.println("C");
}
else if (!str1.equals(str2) && str1 != str2)
{
System.out.println("D");
}
```

What, if anything, is printed when the code segment is executed?

(A) `A`
(B) `B`
(C) `C`
(D) `D`

59. Consider the following class definition.

```
public class Backyard
{
```

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```
    private int length;
    private int width;

    public Backyard(int l, int w)
    {
        length = l;
        width = w;
    }

    public int getLength()
    {
        return length;
    }

    public int getWidth()
    {
        return width;
    }

    public boolean equals(Object other)
    {
        if (other == null)
        {
          return false;
        }

        Backyard b = (Backyard) other;
        return (length == b.getLength() && width == b.getWidth());
    }
}
```

The following code segment appears in a class other than `Backyard`. It is intended to print `true` if `b1` and `b2` have the same lengths and widths, and to print `false` otherwise. Assume that `x`, `y`, `j`, and `k` are properly declared and initialized variables of type `int`.

```
Backyard b1 = new Backyard(x, y);
Backyard b2 = new Backyard(j, k);
System.out.println( /* missing code */ );
```

Which of the following can be used as a replacement for `/* missing code */` so the code segment works as intended?

(A) `b1.equals(b2)`

(B) `equals(b1, b2)`

(C) `b1.equals(b2.getLength(), b2.getWidth())`

(D) `b1.length == b2.length && b1.width == b2.width`

60. In the following Boolean expressions, `x` and `y` are properly declared and initialized `boolean` variables.

Expression A

x && y

Expression B

!x && !y

Which of the following best describes the relationship between values produced by Expression A and Expression B?

(A) Expression A and Expression B evaluate to different values for all values of `x` and `y`.
(B) Expression A and Expression B evaluate to the same value for all values of `x` and `y`.
(C) Expression A and Expression B evaluate to the same value only when `x` and `y` have the same value.
(D) Expression A and Expression B evaluate to the same value only when `x` and `y` have different values.

61. Assume `obj1` and `obj2` are object references. Which of the following best describes when the expression `obj1 == obj2` is true?

(A) When `obj1` and `obj2` are defined within the same method
(B) When `obj1` and `obj2` are instances of the same class
(C) When `obj1` and `obj2` refer to objects that contain the same data
(D) When `obj1` and `obj2` refer to the same object

62. Assume that `x` and `y` are `boolean` variables and have been properly initialized.

```
(x && y) || !(x && y)
```

The result of evaluating the expression above is best described as

(A) always `true`
(B) always `false`
(C) `true` only when `x` is `true` and `y` is `true`
(D) `true` only when `x` and `y` have the same value

63. In the following code segment, `x` is a properly declared and initialized boolean variable and `y` is a properly declared and initialized int variable.

```
boolean a = !x && y > 50000;
boolean b = !(x || y <= 50000);
boolean c = !x || y <= 50000;
boolean d = !(x && y <= 50000);

if (a == b)
```

```
{
  System.out.println("First");
}
if (a == c)
{
  System.out.println("Second");
}
if (b == c)
{
  System.out.println("Third");
}
if (c == d)
{
  System.out.println("Fourth");
}
```

Which of the following is always printed as a result of executing this code segment?

(A)
First

(B)
Second

(C)
Third

(D)
Fourth

64. Consider the following code segment.

```
String alpha = new String("APCS");
String beta = new String("APCS");
String delta = alpha;
System.out.println(alpha.equals(beta));
System.out.println(alpha == beta);
System.out.println(alpha == delta);
```

What is printed as a result of executing the code segment?

(A)
```
false
false
true
```

(B)
```
true
false
false
```

(C)
```
true
false
true
```

(D)
```
true
true
true
```

65. Assume that object references `one`, `two`, and `three` have been declared and instantiated to be of the same type. Assume also that `one == two` evaluates to `true` and that `two.equals(three)` evaluates to `false`. Consider the following code segment.

```
if (one.equals(two))
{
System.out.println("one dot equals two");
}
if (one.equals(three))
{
System.out.println("one dot equals three");
}
if (two == three)
{
System.out.println("two equals equals three");
}
```

What, if anything, is printed as a result of executing the code segment?

(A)
```
one dot equals two
```

(B)
```
one dot equals two
one dot equals three
```

(C)
```
one dot equals three
two equals equals three
```

(D) Nothing is printed.

66. Assume that a, b, and c are boolean variables that have been properly declared and initialized. Which of the following boolean expressions is equivalent to !(a && b) || c?

(A)

a || b || c

(B)

!a && !b || c

(C)

!a && !b && c

(D)

!a || !b || c

67. Assume that `a`, `b`, `c`, and `d` have been declared and initialized with `int` values.

```
!((a >= b) && !(c < d))
```

Which of the following is equivalent to the expression above?

(A)
```
(a < b) || (c < d)
```

(B)
```
(a < b) || (c >= d)
```

(C)
```
(a < b) && (c < d)
```

(D)
```
(a >= b) || (c < d)
```

68. An image on a computer is made up of many pixels. Each pixel in the image contains varying amounts of red, green, and blue. A "mainly red" pixel is defined to be one in which the percentage of red is more than 50, and the percentages of green and blue are each less than 10.

Assume that redPct, greenPct, and bluePct are int variables that have been properly declared and initialized with values representing the percentages of red, green, and blue, respectively, in a pixel.

Which of the following expressions will evaluate to true for a "mainly red" pixel?

(A)

(redPct <= 50) || (greenPct >= 10) || (bluePct >= 10)

(B)

(redPct > 50) && (greenPct < 10) || (bluePct < 10)

(C)

!((redPct <= 50) || (greenPct >= 10) || (bluePct >= 10))

(D)

!(redPct > 50) && !(greenPct > 10) && !(bluePct > 10)

69. Assume that `a` and `b` are variables of type `int`. The expression

```
!(a < b) && !(a > b)
```

is equivalent to which of the following?

(A) `true`
(B) `false`
(C) `a == b`
(D) `a != b`

70. Consider the following class definition.

```
public class Time
{
private int hours;
private int minutes;

public Time(int h, int m)
{
   hours = h;
   minutes = m;
}
/** Returns true if hours * 60 + minutes of this
 *    object is equal to hours * 60 + minutes of object
 *   other; otherwise returns false
 */
public boolean equals(Object other)
  {
    //implementation not shown
 }
}
```

The following code segment appears in a class other than `Time`.

```
Time t1 = new Time(1, 10);
Time t2 = new Time(0, 70);
```

Which of the following statements will print `true` ?

I. `System.out.println(t1 == t2);`
II. `System.out.println(t1.equals(t2));`
III. `System.out.println(equals(t1, t2);`

(A) I only
(B) II only
(C) III only
(D) I and II

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71. Consider the following code segment. The code is intended to read nonnegative numbers and compute their product until a negative number is read; however, it does not work as intended. (Assume that the `readInt` method correctly reads the next number from the input stream.)

```
int k = 0;
int prod = 1;

while (k >= 0)
{
  System.out.print("enter a number: ");
  k = readInt(); // readInt reads the next number from input
  prod = prod * k;
}

System.out.println("product: " + prod);
```

Which of the following best describes the error in the program?

(A) The variable `prod` is incorrectly initialized.

(B) The while condition always evaluates to `true`.

(C) The negative number entered to signal no more input is included in the product.

(D) If the user enters a zero, the computation of the product will be terminated prematurely.

72. Consider the following code segment.

```
int count = 5;
while (count < 100)
{
  count = count * 2;
}
count = count + 1;
```

What is the value of `count` after executing the code segment?

(A)
100

(B)
101

(C)
160

(D)
161

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73. Consider the following code segment.

```
int value = 15;
while (value < 28)
{
  System.out.println(value);
  value++;
}
```

What are the first and last numbers output by the code segment?

(A)

First	Last
15	27

(B)

First	Last
15	28

(C)

First	Last
16	27

(D)

First	Last
16	28

74. Consider the following code segment.

```
int count = 0;
for (int k = 0; k < 10; k++)
{
 count++;
}
System.out.println(count);
```

Which of the following code segments will produce the same output as the code segment above?

(A)

```
int count = 0;
for (int k = 1; k < 10; k++)
{
 count++;
}
System.out.println(count);
```

(B)
```
int count = 1;
for (int k = 1; k <= 10; k++)
{
 count++;
}
System.out.println(count);
```

(C)
```
int count = 1;
for (int k = 0; k <= 9; k++)
{
 count++;
}
System.out.println(count);
```

(D)
```
int count = 0;
for (int k = 9; k >= 0; k--)
{
 count++;
}
System.out.println(count);
```

75. In the following code segment, `numOne` and `numTwo` are properly declared and initialized `int` variables.

```
while (numOne > 1)
{
  if (numOne % numTwo == 0)
  {
    numOne = numOne / numTwo;
  }
}
```

Which of the following values for `numOne` and `numTwo` causes an infinite loop to occur when the code segment is executed?

(A) `numOne = 16` and `numTwo = 2`
(B) `numOne = 100` and `numTwo = 10`
(C) `numOne = 125` and `numTwo = 5`
(D) `numOne = 128` and `numTwo = 4`

76. Which of the following code segments produces the output "987654321"?

(A)

```
int num = 10;
while (num > 0)
{
  System.out.print(num);
  num--;
}
```

(B)

```
int num = 10;
while (num > 1)
{
  System.out.print(num);
  num--;
}
```

(C)

```
int num = 10;
while (num > 1)
{
  num--;
  System.out.print(num);
}
```

(D)

```
int num = 10;
while (num >= 1)
{
  num--;
  System.out.print(num);
}
```

77. Consider the following code segment.

```
int k = 0;
/* missing loop header */
{
 k++;
```

```
    System.out.print(k + " ");
  }
```

Which of the following can be used as a replacement for `/* missing loop header */` so that the code segment prints out the string `"1 2 3 4 "`?

(A) `while (k < 4)`
(B) `while (k < 5)`
(C) `while (k <= 4)`
(D) `while (k <= 5)`

78. Consider the following code segment.

```
int num = 1;
while (num < 5)
{
System.out.print("A");
num += 2;
}
```

What is printed as a result of executing the code segment?

(A) `A`
(B) `AA`
(C) `AAA`
(D) `AAAA`

79. The following code segment is intended to print the number of digits in the `int` variable `num`, where `num > 0`.

```
int count = 0;
while ( /* missing condition */ )
{
  count++;
  num = num / 10;
}
System.out.println(count);
```

Which of the following can be used to replace `/* missing condition */` so that the code segment works as intended?

(A)
count != 0

(B)

count > 0

(C)

num >= 0

(D)

num != 0

80. In the following code segment, x is an int variable with a positive value.

```
int temp = x;
while (temp > 0)
{
  temp -= 2;
}
System.out.println(temp == 0);
```

Which of the following best describes the behavior of the code segment?

(A) It prints 0 if x is even and prints -1 otherwise.
(B) It prints -1 if x is even and prints 0 otherwise.
(C) It prints true if x is even and prints false otherwise.
(D) It prints false if x is even and prints true otherwise.

81. Consider the following code segment.

```
int a = 1;
String result = "";
while (a < 20)
{
  result += a;
  a += 5;
}
System.out.println(result);
```

What is printed as a result of executing this code segment?

(A)

21

(B)

161116

(C)

161161

(D)

16111621

82. Which of the following code segments can be used to print only the even integers between 2 and 10, inclusive?

(A)

```
int num = 0;
while(num < 10)
{
  System.out.println(num);
  num += 2;
}
```

(B)

```
int num = 0;
while(num <= 10)
{
  System.out.println(num);
  num += 2;
}
```

(C)

```
int num = 0;
while(num < 10)
{
  num += 2;
  System.out.println(num);
}
```

(D)

```
int num = 0;
while(num <= 10)
{
  num += 2;
  System.out.println(num);
}
```

83. Consider the following code segment.

```
int j = 0;
int sum = 0;

while (j <= 5)
{
  j++;
  sum += j;
}
```

Which of the following code segments assigns the same value to sum as the preceding code segment?

(A)

```
int sum = 0;
for (int j = 0; j <= 5; j++)
{
  j++;
  sum += j;
}
```

(B)

```
int sum = 0;
for (int j = 0; j <= 6; j++)
{
  j++;
  sum += j;
}
```

(C)

```
int sum = 0;
for (int j = 1; j <= 5; j++)
{
  sum += j;
}
```

(D)

```
int sum = 0;
for (int j = 1; j <= 6; j++)
{
  sum += j;
}
```

84. Consider the following code segment.

```
int val = 48;
int div = 6;
while ((val % 2 == 0) && div > 0)
{
if (val % div == 0)
{
    System.out.print(val + " ");
}
val /= 2;
div--;
}
```

What is printed when the code segment is executed?

(A) `48 12 6`
(B) `48 12 6 3`
(C) `48 12 6 3 1`
(D) `48 24 12 6`

85. Consider the following code segment, which is intended to print the sum of all the odd integers from 1 to 101, inclusive.

```
int r = 1;
int sum = 0;

while (r <= 101)
{
```

```
  /* missing code */
  r++;
}
System.out.println(sum);
```

Which of the following can be used to replace `/* missing code */` so that the code segment works as intended?

(A)

```
if (r % 2 != 1)
{
  sum += r;
}
```

(B)

```
if (r % 2 != 1)
{
  sum++;
}
```

(C)

```
if (r % 2 == 1)
{
  sum += r;
}
```

(D)

```
if (r % 2 == 1)
{
  sum++;
}
```

86. Consider the following method definition. The method `printAllCharacters` is intended to print out every character in `str`, starting with the character at index `0`.

```
public static void printAllCharacters(String str)
{
for (int x = 0; x < str.length(); x++)   // Line 3
{
    System.out.print(str.substring(x, x + 1));
}
```

```
}
```

The following statement is found in the same class as the `printAllCharacters` method.

```
printAllCharacters("ABCDEFG");
```

Which choice best describes the difference, if any, in the behavior of this statement that will result from changing `x < str.length()` to `x <= str.length()` in line 3 of the method?

(A) The method call will print fewer characters than it did before the change because the loop will iterate fewer times.

(B) The method call will print more characters than it did before the change because the loop will iterate more times.

(C) The method call, which worked correctly before the change, will now cause a run-time error because it attempts to access a character at index `7` in a string whose last element is at index `6`.

(D) The method call, which worked correctly before the change, will now cause a run-time error because it attempts to access a character at index `8` in a string whose last element is at index `7`.

87. Consider the following method.

```
/** Precondition: bound >= 0 */
public int sum(int bound)
{
 int answer = 0;
 for (int i = 0; i < bound; i++)
 {
    answer += bound;
 }
 return answer;
}
```

Assume that `sum` is called with a parameter that satisfies the precondition and that it executes without error. How many times is the test expression `i < bound` in the `for` loop header evaluated?

(A) `bound - 1`

(B) `bound`

(C) `bound + 1`

(D) An unknown number of times

88. Consider the following two code segments. Code segment 2 is a revision of code segment 1 in which the loop header has been changed.

Code Segment 1

```
for (int k = 1; k <= 5; k++)
{
  System.out.print(k);
}
```

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Code Segment 2

```
for (int k = 5; k >= 1; k--)
{
  System.out.print(k);
}
```

Which of the following best explains the difference, if any, between the output of code segment 1 and code segment 2?

(A) Both code segments produce the same output, because they both iterate four times.
(B) Both code segments produce the same output, because they both iterate five times.
(C) Code segment 1 prints more values than code segment 2 does, because it iterates for one additional value of k.
(D) The code segments print the same values but in a different order, because code segment 1 iterates from 1 to 5 and code segment 2 iterates from 5 to 1.

89. Consider the following code segment.

```
String word = "computer";
int num = 3;
String result = "";

for (int k = num; k >= 0; k--)
{
   result += word.substring(0, k);
}

System.out.print(result);
```

What is printed as a result of executing the code segment?

(A) `comcoc`
(B) `ccocomcomp`
(C) `compcomcoc`
(D) `comcomcomcom`

90. Consider the following code segment.

```
for (int j = 1; j < 10; j += 2)
{
  System.out.print(j);
}
```

Which of the following code segments will produce the same output as the given code segment?

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(A)

```
int j = 1;
while (j < 10)
{
  j += 2;
  System.out.print(j);
}
```

(B)

```
int j = 1;
while (j < 10)
{
  System.out.print(j);
  j += 2;
}
```

(C)

```
int j = 1;
while (j <= 10)
{
  j += 2;
  System.out.print(j);
}
```

(D)

```
int j = 1;
while (j >= 10)
{
  j += 2;
  System.out.print(j);
}
```

91. Consider the following code segment.

```
for (int k = 1; k <= 7; k += 2)
{
 System.out.print(k);
```

```
}
```

Which of the following code segments will produce the same output as the code segment above?

(A)
```
for (int k = 0; k < 7; k += 2)
{
 System.out.print(k);
}
```

(B)
```
for (int k = 0; k <= 8; k += 2)
{
 System.out.print(k + 1);
}
```

(C)
```
for (int k = 1; k < 7; k += 2)
{
 System.out.print(k + 1);
}
```

(D)
```
for (int k = 1; k <= 8; k += 2)
{
 System.out.print(k);
}
```

92. Consider the following two code segments, which are intended to produce identical outputs.

Code Segment 1

```
for (int i = 0; i < 10; i++)
{
  System.out.println("counting " + i);
}
```

Code Segment 2

```
int x = 0;
while (x < 10)
{
  x = x + 1;
  System.out.println("counting " + x);
}
```

Which of the following changes can be made to Code Segment 2 so that the outputs of the two code segments are identical as intended?

(A) Move the statement `x = x + 1;` so that it follows the print statement inside the loop body.
(B) Rename the variable `x` to be `i` so that it is the same variable name as in Code Segment 1.
(C) Remove the statement `x = x + 1;` because this statement is not in the loop body in Code Segment 1.

(D) Change the statement x = x + 1; to be x++; so that the increment of the loop control variable is the same in both code segments.

93. Consider the following code segment.

```
for (int j = 10; j >= 0; j -= 2)  // Line 1
{                                 // Line 2
  System.out.print(j – 1);        // Line 3
}                                 // Line 4
```

This code segment is intended to produce only the following output. It does not work as intended.

97531

Which of the following changes can be made so that this code segment works as intended?

(A) In line 1, changing the condition j >= 0 to j > 0
(B) In line 1, changing the initialization j = 10 to j = 9
(C) In line 3, changing the expression j - 1 to j + 1
(D) In line 3, changing the expression j - 1 to j - 2

94. Consider the following code segment.

```
for (int m = 16; m > 0; m -= 2)
{
  if ((m % 3) == 1)
  {
    System.out.print(m + " ");
  }
}
```

What is printed as a result of executing this code segment?

(A)
15 12 9 6 3

(B)
12 6 0

(C)
16 14 12 10 8 6 4 2

(D)
16 10 4

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95. Consider the following code segment.

```
int sum = 0;
for (int j = 1; /* missing condition */; j += 2)
{
  sum += j;
}
```

Which of the following replacements for /* missing condition */ will cause an infinite loop?

(A)
j == 100

(B)
j != 100

(C)
j <= 100

(D)
j >= 100

96. Consider the following code segment.

```
for (int k = 0; k < 20; k = k + 2)
{
  if (k % 3 == 1)
  {
    System.out.print(k + "  ");
  }
}
```

What is printed as a result of executing the code segment?

(A) 4 10 16
(B) 0 6 12 18
(C) 1 4 7 10 13 16 19
(D) 0 2 4 6 8 10 12 14 16 18

97. Consider the following code segment.

```
int sum = 0;
for (int k = 1; k <= 30; k = k + 2)
{
```

```
  sum += k;
}
System.out.println(sum);
```

Which of the following best describes the output of the code segment?

(A) The segment will print the sum of only the even integers from 1 through 30, inclusive, because it starts `sum` at `0`, increments `k` by `2`, and terminates when `k` exceeds `30`.

(B) The segment will print the sum of only the odd integers from 1 through 30, inclusive, because it starts `k` at `1`, increments `k` by `2`, and terminates when `k` exceeds `30`.

(C) The segment will print the sum of only the even integers from 1 through 60, inclusive, because it starts `sum` at `0`, increments `k` by `2`, and iterates 30 times.

(D) The segment will print the sum of only the odd integers from 1 through 60, inclusive, because it starts `k` at `1`, increments `k` by `2`, and iterates 30 times.

98. Consider the following code segment.

```
int num = 1;
for (int k = 2; k < 10; k++)
{
num = 0;
num = num + k;
}
```

What will be the value of `num` after the loop is executed?

(A) `9`

(B) `10`

(C) `44`

(D) `45`

99. Consider the following code segment.

```
for (int j = 0; j < 4; j++)
{
  for (/* missing code */)
  {
    System.out.print(j + " ");
  }
  System.out.println();
}
```

This code segment is intended to produce the following output.

```
0
1 1
2 2 2
3 3 3 3
```

Which of the following can be used to replace /* missing code */ so that this code segment works as intended?

(A)

```
int k = 0; k < j; k++
```

(B)

```
int k = 0; k <= j; k++
```

(C)

```
int k = j; k < 4; k++
```

(D)

```
int k = j; k <= 4; k++
```

100. Consider the following code segment.

```
for (int j = 4; j > 0; j--)
{
  for (/* missing code */)
  {
    System.out.print(k + " ");
  }
  System.out.println();
}
```

This code segment is intended to produce the following output.

```
0 1 2 3
0 1 2
0 1
0
```

Which of the following can be used to replace /* missing code */ so that this code segment works as intended?

(A)

```
int k = 0; k < j; k++
```

(B)

```
int k = 0; k <= j; k++
```

(C)

```
int k = j; k > 0; k--
```

(D)

```
int k = j; k >= 0; k--
```

101. Consider the following method.

```
/** Precondition: num > 0 */
public static int doWhat(int num)
{
  int var = 0;
  for (int loop = 1; loop <= num; loop = loop + 2)
  {
    var += loop;
  }
  return var;
}
```

Which of the following best describes the value, if any, returned from a call to doWhat?

(A) The sum of all integers between 1 and num, inclusive
(B) The sum of all even integers between 1 and num, inclusive
(C) The sum of all odd integers between 1 and num, inclusive
(D) No value is returned because of an infinite loop.

102. In the following code segment, num has been assigned a positive int value.

```
int x = 0;
int temp = num;

while (temp > 0)
{
  x += temp % 10;
  temp /= 10;
}

System.out.println(x);
```

Which of the following best describes the value printed by this code segment?

(A) A count of the number of digits in num
(B) The average of the digits in num
(C) The sum of the digits in num
(D) Whether num is evenly divisible by 10

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103. Consider the following code segment.

```
int a = 24;
int b = 30;
while (b != 0)
{
  int r = a % b;
  a = b;
  b = r;
}

System.out.println(a);
```

What is printed as a result of executing the code segment?

(A) 0
(B) 6
(C) 24
(D) 30

104. In the following code segment, n is a properly declared and initialized int variable whose value is positive.

```
while (n > 0)        // Line 1
{                    // Line 2
  n /= 10;           // Line 3
}                    // Line 4
System.out.println(n);  // Line 5
```

This code segment is intended to print the leftmost digit of n. For example, if n is 302, this code segment should print 3. However, this code segment is not working as intended. Which of the following changes can be made so that this code segment works as intended?

(A) In line 1, changing the condition n > 0 to n / 10 > 0
(B) In line 1, changing the condition n > 0 to n % 10 > 0
(C) In line 3, changing the statement n /= 10 to n -= 10
(D) In line 3, changing the statement n /= 10 to n %= 10

105. In the following code segment, the variable num is assigned a random value inside a loop.

```
int result = 0;
for (int j = 1; j <= 20; j++)
{
   int num = ((int) Math.random() * 10) + 1;
   if (num > result)
   {
      result = num;
```

```
    }
}
```

Which of the following best describes the value of the variable `result` after executing the code segment?

(A) The first random value assigned to `num`
(B) The last random value assigned to `num`
(C) The least random value assigned to `num`
(D) The greatest random value assigned to `num`

106. In the following code segment, assume that low and high are properly declared and initialized int variables and that low < high. The code segment is intended to print the sum of the integers between low and high, inclusive, but does not always work as intended.

```
int sum = 0;        // line 1
int j = low;        // line 2
while (sum <= high)  // line 3
{
  sum += j;         // line 5
  j++;              // line 6
}
System.out.println(sum);
```

Which of the following changes can be made so that this code segment works as intended?

(A) Changing line 1 to int sum = low;
(B) Changing line 2 to int j = 0;
(C) Changing line 3 to while (j <= high)
(D) Interchanging lines 5 and 6

107. Consider the following code segment.

```
int num = 245;
int temp = num;

while (temp > 0)
{
  System.out.print(temp % 10 + " ");
  temp /= 10;
}
```

What is printed as a result of executing this code segment?

(A)
2 4 5

(B)

4 5

(C)

5 4

(D)

5 4 2

108. Consider the following code segment.

```
int x = 1;
while (x < 10)
{
  if (x % 2 == 0)
  {
    System.out.print(x + " ");
  }
  /* missing code */
}
```

Which of the following replacements for /* missing code */ will cause the code segment to print nothing?

(A)

x++;

(B)

x += 2;

(C)

x += 3;

(D)

x *= 2;

109. Consider the following code segment.

```
int k = 1;
while (k < 20)
{
  if ((k % 3) == 1)
    System.out.print(k + " ");

  k++;
}
```

What is printed as a result of executing this code segment?

(A) `2 5 8 11 14 17`

(B) `3 6 9 12 15 18`

(C) `1 4 7 10 13 16 19`

(D) `1 3 5 7 9 11 13 15 17 19`

110. An acronym is formed by extracting the first letter of each word in a phrase of one or more words. For example, the acronym of the phrase "Laugh Out Loud" is "LOL", and the acronym of the phrase "If you know you know" is "Iykyk."

A phrase must meet the following conditions.

- The phrase has exactly one space between each word.
- The phrase does not have leading or trailing spaces.
- The final word in the phrase has at least two characters.

In the following code segment, the String variable phrase has been properly declared and initialized such that it meets these conditions. The code segment is intended to create the acronym of the String stored in phrase and store it in the variable text.

```
String temp = phrase;
String text = temp.substring(0, 1);
int i = temp.indexOf(" ");
while (i > 0)
{
  /* missing code */
}
```

Which of the following can be used to replace /* missing code */ so that the code segment works as intended?

(A)

```
text += temp.substring(0, 1);
temp = temp.substring(i + 1);
i = temp.indexOf(" ");
```

(B)

```
temp = temp.substring(i + 1);
text += temp.substring(0, 1);
i = temp.indexOf(" ");
```

(C)

```
text += temp.substring(i, i + 1);
i = i + 1;
```

(D)

```
text += temp.substring(i, i + 1);
i = temp.indexOf(" ");
```

111. Consider the following method.

```
public static String changeStr(String str)
{
  String result = "";
  for (int i = str.length() - 1; i >= str.length() / 2; i -= 2)
  {
    result += str.substring(i, i + 1);
  }
  return result;
}
```

What value is returned as a result of the method call `changeStr("12345")`?

(A)

"4"

(B)

"53"

(C)

"531"

(D)

"543"

112. In the following code segment, str is a properly declared and initialized string. The code segment is intended to count the number of times that "a" appears in str.

```
int count = 0;
String temp = str;
int loc = temp.indexOf("a");

while (loc >= 0)
{
```

```
  /* missing code */
}
```

Which of the following can be used to replace /* missing code */ so that the code segment works as intended?

(A)

```
count++;
temp = str.substring(loc + 1);
```

(B)

```
count++;
loc++;
temp = str.substring(loc);
```

(C)

```
count++;
temp = temp.substring(loc);
loc = temp.indexOf("a");
```

(D)

```
count++;
temp = temp.substring(loc + 1);
loc = temp.indexOf("a");
```

113. Consider the following code segment.

```
String str = "ABCDEF";
String result = "";
/* missing code */
System.out.println(result);
```

Which of the following can be used to replace /* missing code */ so that this code segment prints the string "EFCDAB"?

(A)

```
for (int j = str.length() - 1; j >= 0; j -= 1)
{
  result += str.substring(j, j + 1);
}
```

(B)

```
for (int j = str.length() - 1; j >= 0; j -= 2)
{
  result += str.substring(j, j + 2);
}
```

(C)

```
for (int j = str.length() - 2; j >= 0; j -= 1)
{
  result += str.substring(j, j + 2);
}
```

(D)

```
for (int j = str.length() - 2; j >= 0; j -= 2)
{
  result += str.substring(j, j + 2);
}
```

114. Consider the following method.

```
public static String mystery(String input)
{
  String output = "";
  for (int k = 1; k < input.length(); k = k + 2)
  {
    output += input.substring(k, k + 1);
  }
  return output;
}
```

What is returned as a result of the call mystery("computer")?

(A)

"computer"

(B)

"cmue"

(C)

"optr"

(D)

"ompute"

115. Consider the following code segment.

```
String str = "qrstu";
String result = "";
for (int j = 0; j < str.length(); j++)
{
  result += str.substring(0, j + 1);
}
System.out.println(result);
```

What, if anything, is printed as a result of executing this code segment?

(A)

rrsrstrstu

(B)

qqrqrsqrst

(C)

qqrqrsqrstqrstu

(D) Nothing is printed because an IndexOutOfBoundsException occurs.

116. Consider the following code segment.

```
int j = 1;
while (j < 5)
{
 int k = 1;
 while (k < 5)
 {
     System.out.println(k);
     k++;
 }
 j++;
}
```

Which of the following best explains the effect, if any, of changing the first line of code to `int j = 0;` ?

(A) There will be one more value printed because the outer loop will iterate one additional time.
(B) There will be four more values printed because the outer loop will iterate one additional time.
(C) There will be one less value printed because the outer loop will iterate one fewer time.

(D) There will be no change to the output of the code segment.

117. Consider the following code segment.

```
for (int x = 0; x <= 4; x++)      // Line 1
{
for (int y = 0; y < 4; y++)    // Line 3
{
    System.out.print("a");
}
System.out.println();
}
```

Which of the following best explains the effect of simultaneously changing `x <= 4` to `x < 4` in line 1 and `y < 4` to `y <= 4` in line 3 ?

(A) `"a"` will be printed fewer times because while each output line will have the same length as before, the number of lines printed will decrease by 1.

(B) `"a"` will be printed more times because while the number of output lines will be the same as before, the length of each output line will increase by 1.

(C) `"a"` will be printed the same number of times because while the number of output lines will decrease by 1, the length of each line will increase by 1.

(D) The output of the code segment will not change in any way.

118. Consider the following code segment.

```
for (int outer = 0; outer < 3; outer++)
{
  for ( /* missing loop header */ )
  {
    System.out.print(outer + "" + inner + "_");
  }
}
```

Which of the following can be used as a replacement for `/* missing loop header */` so that the code segment produces the output `00_01_02_11_12_22_`?

(A)

```
int inner = 0; inner < 3; inner++
```

(B)

```
int inner = 1; inner < 3; inner++
```

(C)

```
int inner = outer - 1; inner < 3; inner++
```

(D)

int inner = outer; inner < 3; inner++

119. Consider the following output.

```
1   1   1   1   1
2   2   2   2
3   3   3
4   4
5
```

Which of the following code segments will produce this output?

(A)
```
for (int j = 1; j <= 5; j++)
{
  for (int k = 1; k <= j; k++)
  {
    System.out.print(j + "  ");
  }
  System.out.println();
}
```

(B)
```
for (int j = 1; j <= 5; j++)
{
  for (int k = 5; k >= 1; k--)
  {
    System.out.print(j + "  ");
  }
  System.out.println();
}
```

(C)
```
for (int j = 1; j <= 5; j++)
{
  for (int k = 5; k >= j; k--)
  {
    System.out.print(j + "  ");
  }
  System.out.println();
}
```

(D)
```
for (int j = 1; j <= 5; j++)
{
  for (int k = j; k <= 5; k++)
  {
    System.out.print(k + "  ");
  }
  System.out.println();
}
```

120. Consider the following code segment.

```
int count = 0;
for (int x = 1; x <= 3; x++)
{
  /* missing loop header */
  {
    count++;
  }
}
System.out.println(count);
```

Which of the following can be used to replace `/* missing loop header */` so that the code segment prints `6`?

(A)

```
for (int y = 0; y <= 2; y++)
```

(B)

```
for (int y = 0; y < 3; y++)
```

(C)

```
for (int y = 3; y <= x; y++)
```

(D)

```
for (int y = 0; y < x; y++)
```

121. Consider the following code segment.

```
int a = 1;
while (a <= 2)
{
  int c = 1;
  while (/* missing condition */)
  {
    System.out.print("*");
    c++;
  }
  a++;
}
```

The code segment is intended to print `"******"`.

Which of the following can be used to replace `/* missing condition */` so that the code segment works as intended?

(A)

```
c > 2
```

(B)

c < 3

(C)

c <= 3

(D)

c >= 3

122. Consider the following code segment.

```
for (int outer = 1; outer <= 4; outer++)
{
  for (int inner = outer; inner <= 4; inner++)
  {
    System.out.print(inner + " ");
  }
  System.out.println();
}
```

What is printed as a result of executing the code segment?

(A)

1 1 1 1

2 2 2

3 3

4

(B)

1 1 1 1

2 2 2 2

3 3 3 3

4 4 4 4

(C)

1 2 3 4

2 3 4

3 4

4

(D)

1 2 3 4

1 2 3 4

1 2 3 4

1 2 3 4

123. Consider the following code segment.

```
/* missing loop header */
{
  for (int k = 0; k < 4; k++)
  {
    System.out.print(k);
  }
  System.out.println();
}
```

This code segment is intended to produce the following output.

```
0123
0123
0123
```

Which of the following can be used to replace /* missing loop header */ so that the code segment works as intended?

(A)

```
for (int j = 0; j < 2; j++)
```

(B)

```
for (int j = 1; j <= 2; j++)
```

(C)

```
for (int j = 0; j <= 6; j += 2)
```

(D)

```
for (int j = 1; j < 6; j += 2)
```

124. Consider the following output.

```
1
1 2
1 2 3
1 2 3 4
```

```
1 2 3 4 5
1 2 3 4 5 6
```

Which of the following code segments will produce the output shown above?

(A)
```
for (int j = 1; j <= 6; j++)
{
  for (int k = 1; k < j; k++)
    System.out.print(" " + k);
  System.out.println();
}
```

(B)
```
for (int j = 1; j <= 6; j++)
{
  for (int k = 1; k <= j; k++)
    System.out.print(" " + j);
  System.out.println();
}
```

(C)
```
for (int j = 1; j <= 6; j++)
{
  for (int k = 1; k <= j; k++)
    System.out.print(" " + k);
  System.out.println();
}
```

(D)
```
for (int j = 1; j < 6; j++)
{
  for (int k = 1; k <= j; k++)
    System.out.print(" " + k);
  System.out.println();
}
```

125. Consider the following code segment.

```
for (int j = 0; j < 3; j++)
{
for (int k = 0; k < 4; k++)
{
    System.out.println("Fun");
}
}
```

Which of the following best explains how changing the outer `for` loop header to `for (int j = 0; j <= 3; j++)` affects the output of the code segment?

(A) The output of the code segment will be unchanged.

(B) The string `"Fun"` will be printed more times because the outer loop will execute more times.

(C) The string `"Fun"` will be printed more times because the inner loop will execute more times in each iteration of the outer loop.

(D) The string `"Fun"` will be printed fewer times because the outer loop will execute fewer times.

126. Consider the following code segment.

```
for (int j = 1; j <= 4; j++)       // Line 1
{
  for (int k = j + 1; k >= 1; k--)  // Line 3
  {
    System.out.print(j + " ");     // Line 5
  }
  System.out.println();
}
```

The code segment is intended to produce the following output, but does not work as intended.

```
1
2 2
3 3 3
4 4 4 4
```

Which of the following changes can be made so that the code segment works as intended?

(A) Changing line 1 to `for (int j = 1; j < 4; j++)`
(B) Changing line 3 to `for (int k = 1; k <= j + 1; k++)`
(C) Changing line 3 to `for (int k = j; k >= 1; k--)`
(D) Changing line 5 to `System.out.print(k + " ");`

127. Consider the following code segment.

```
for (int outer = 0; outer < n; outer++)
{
  for (int inner = 0; inner <= outer; inner++)
  {
    System.out.print(outer + " ");
  }
}
```

If `n` has been declared as an integer with the value 4, what is printed as a result of executing the code segment?

(A) `0 1 2 3`
(B) `0 1 2 2 3 3 3`
(C) `0 1 1 2 2 2 3 3 3 3`
(D) `0 0 1 0 1 2 0 1 2 3`

128. Consider the following method.

```
public int sol(int lim)
{
```

```
    int s = 0;

    for (int outer = 1; outer <= lim; outer++)
    {
      for (int inner = outer; inner <= lim; inner++)
      {
        s++;
      }
    }

    return s;
  }
```

What value is returned as a result of the call `sol(10)`?

(A) 20
(B) 45
(C) 55
(D) 100

129. The following code segment appears in a method in the same class as the method doSomething().

```
int i = 1;
while (i < 3)
{
   for (int j = 1; j <= 5; j++)
   {
      doSomething();
   }
   i++;
}
```

How many times will doSomething() be called as a result of executing this code segment?

(A) 5
(B) 8
(C) 10
(D) 15

130. Consider the following code segment.

```
for (int num = 0; num < 10; num += 2)
{
for (int val = 0; val < 5; val++)
{
```

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```
            System.out.println("hop");
        }
        }
```

How many times will `System.out.println("hop")` be executed?

(A) 5
(B) 10
(C) 25
(D) 50

131. Consider the following code segment.

```
int j = 1;
while (j < 4)
{
  for (int k = 0; k <= 4; k++)
  {
    System.out.println("hello");    // line 6
  }
  j++;
}
```

How many times will the statement in line 6 be executed as a result of running this code segment?

(A) 12
(B) 15
(C) 16
(D) 20

132. Consider the following code segment.

```
for (int outer = 0; outer < 5; outer++)
{
  for (int inner = 4; inner >= 0; inner--)
  {
    System.out.println("*");            // Line 5
  }
}
```

How many times will the statement in line 5 be executed as a result of running this code segment?

(A) 20
(B) 24
(C) 25
(D) 30

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133. Consider the following code segment.

```
for (int x = 1; x <= 3; x++)
{
   for (int y = 5; y > 1; y--)
   {
      System.out.print("*");
   }
}
```

Which of the following best describes the behavior of this code segment?

(A) The print statement is executed 7 times because the outer loop iterates 3 times and the inner loop iterates 4 times.
(B) The print statement is executed 8 times because the outer loop iterates 3 times and the inner loop iterates 5 times.
(C) The print statement is executed 12 times because the outer loop iterates 3 times and the inner loop iterates 4 times for each iteration of the outer loop.
(D) The print statement is executed 15 times because the outer loop iterates 3 times and the inner loop iterates 5 times for each iteration of the outer loop.

134. Consider the following code segment.

```
int counter = 0;
for (int x = 10; x > 0; x--)
{
  for (int y = x; y <= x; y++)
  {
    counter++;      // Line 6
  }
}
```

How many times will the statement in line 6 be executed as a result of running the code segment?

(A) 1
(B) 10
(C) 11
(D) 20

135. Consider the following code segment.

```
double sum = 0.0;
for (int outer = 1; outer <= 4; outer++)
{
   for (int inner = 0; inner <= 4; inner++)
   {
      sum += Math.random();
   }
}
```

How many times will Math.random() be called as a result of executing this code segment?

(A) 5
(B) 9
(C) 16
(D) 20

136. Consider the following code segment.

```
int val = 1;
while (val <= 6)
{
  for (int k = 0; k <= 2; k++)
  {
    System.out.println("Surprise!");   // Line 6
  }
  val++;
}
```

How many times will the statement in line 6 be executed as a result of running the code segment?

(A) 3
(B) 12
(C) 15
(D) 18

137. In the following code segment, `outerMax` and `innerMax` are properly declared and initialized `int` variables that are greater than or equal to zero.

```
for (int outer = 0; outer < outerMax; outer++)
{
  for (int inner = 0; inner < innerMax; inner++)
  {
    System.out.println("*");
  }
}
```

Which of the following best describes the behavior of the code segment?

(A) It prints `"*"` a total of `outerMax + innerMax` times.
(B) It prints `"*"` a total of `outerMax * innerMax` times.
(C) It prints `"*"` a total of `(outerMax + 1) + (innerMax + 1)` times.
(D) It prints `"*"` a total of `(outerMax + 1) * (innerMax + 1)` times.

138. Consider the following two code segments. Assume that the int variables m and n have been properly declared and initialized and are both greater than 0.

Code Segment 1

```
for (int i = 0; i < m * n; i++)
{
  System.out.print("*");
}
```

Code Segment 2

```
for (int j = 1; j <= m; j++)
{
  for (int k = 1; k < n; k++)
  {
    System.out.print("@");
  }
}
```

Assume that the initial values of m and n are the same in Code Segment 1 as they are in Code Segment 2. Which of the following correctly compares the number of times that "*" and "@" are printed when each code segment is executed?

(A) "*" is printed m fewer times than "@".
(B) "*" is printed n fewer times than "@".
(C) "*" is printed m more times than "@".
(D) "*" is printed n more times than "@".

139. **The question refer to the following code segment.**

```
int k = a random number such that 1 ≤ k ≤ n ;
for (int p = 2; p <= k; p++)
  for (int r = 1; r < k; r++)
    System.out.println("Hello");
```

What is the maximum number of times that `Hello` will be printed?

(A) $n - 1$
(B) $n - 2$
(C) $(n - 1)^2$
(D) n^2

140. **The question refer to the following code segment.**

```
int k = a random number such that 1 ≤ k ≤ n ;
for (int p = 2; p <= k; p++)
  for (int r = 1; r < k; r++)
    System.out.println("Hello");
```

What is the minimum number of times that `Hello` will be printed?

(A) 0
(B) 1
(C) 2

(D) `n - 1`

141. Consider the following code segment.

```
int value = 100;
while (value % 5 == 0)
{
  value /= 2;
}
System.out.println(value);
```

What is printed as a result of executing this code segment?

(A)
12

(B)
13

(C)
25

(D)
100

142. Consider the following expression.

```
(3 + 4 == 5) != (3 + 4 >= 5)
```

What value, if any, does the expression evaluate to?

(A)
true

(B)
false

(C)
5

(D) No value; relational operators cannot be used on arithmetic expressions.

143. Consider the following code segment.

```
int quant = 20;
int unitPrice = 4;
int ship = 8;
int total;
if (quant > 10)
{
  unitPrice = 3;
}
if (quant > 20)
{
  ship = 0;
}
total = quant * unitPrice + ship;
```

What is the value of total after this code segment is executed?

(A) 20

(B) 60

(C) 68

(D) 88

144. Assume that `x` and `y` are variables of type `int`. Which of the following Java expressions never results in a division by zero?

(A) `((y / x) == 0) && (x != 0)`

(B) `((y / x) == 0) || (x != 0)`

(C) `(x != 0) && ((y / x) == 0)`

(D) `(x != 0) || ((y / x) == 0)`

145. Consider the following code segment.

```
boolean b1 = true && ((5 % 2) == 0);
boolean b2 = /* missing expression */ ;
```

Which of the following can replace `/* missing expression */` so that the value stored in `b2` is the same as the value stored in `b1` after the code segment has executed?

(A)
false || ((5 % 2) == 1)

(B)
false && ((5 % 2) == 1)

(C)
true || ((5 % 2) == 1)

(D)
true && ((5 % 2) == 1)

146. Consider the following methods.

```
/** Preconditions: a > 0 and b > 0 */
public static int methodX(int a, int b)
{
  int loopCount = 0;
  for (int i = 0; i < a / b; i++)
  {
    loopCount++;
  }
  return loopCount;
}
```

```
/** Preconditions: a > 0 and b > 0 */
public static int methodY(int a, int b)
{
  int loopCount = 0;
  int i = 0;
  while (i < a)
  {
    loopCount++;
    i += b;
  }
  return loopCount;
}
```

Which of the following best describes the conditions under which methodX and methodY return the same value?

(A) When a and b are both even
(B) When a and b are both odd
(C) When a % b is equal to zero
(D) When a % b is equal to one

147. Consider the following code segment.

```
for (int k = 0; k < 9; k = k + 2)
{
if ((k % 2) != 0)
{
    System.out.print(k + " ");
}
}
```

What, if anything, is printed as a result of executing the code segment?

(A) `0 2 4 6 8`

(B) `1 3 5 7 9`

(C) `1 3 5 7`

(D) Nothing is printed.

148. Consider the following code segment.

```
int num = 1;
int count = 0;
while (num <= 10)
{
  if (num % 2 == 0 && num % 3 == 0)
  {
    count++;
  }
  num++;
}
```

What value is stored in the variable count as a result of executing the code segment?

(A) 1

(B) 3

(C) 5

(D) 7

149. In the following code segment, num has been assigned a positive int value. The following code segment is intended to print true if num is even and is intended to print false otherwise.

```
boolean isEven = true;
if (/* missing code */)
{
  isEven = false;
}
System.out.println(isEven);
```

Which of the following can replace /* missing code */ so that this code segment works as intended?

(A)
```
num / 2 == 0
```

(B)
```
num / 2 != 0
```

(C)
```
num % 2 == 0
```

(D)
```
num % 2 != 0
```

150. Consider the following code segment. Assume that the int variable input has been properly declared and initialized.

```
int answer = 1;
if (input != 0)
{
  int count = 1;

  while (count != input)
  {
    count++;
    answer *= count;
  }
}

System.out.println(answer);
```

Which of the following best describes the condition in which this code segment always results in integer overflow?

(A) When input == 0
(B) When input == 1
(C) When input > 0
(D) When input < 0

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151. Consider the following method.

```
public static int mystery(boolean a, boolean b, boolean c)
{
int answer = 7;

if (!a)
{
   answer += 1;
}

if (b)
{
   answer += 2;
}

if (c)
{
   answer += 4;
}

return answer;
}
```

Which of the following method calls will return the value `11` ?

(A) `mystery(true, true, true)`

(B) `mystery(true, false, true)`

(C) `mystery(false, true, false)`

(D) `mystery(false, false, true)`

152. Consider the following code segment.

```
double regularPrice = 100.0;
boolean onClearance = true;
boolean hasCoupon = false;

double finalPrice = regularPrice;
if (onClearance)
{
  finalPrice -= finalPrice * 0.25;
}
```

```
if (hasCoupon)
{
  finalPrice -= 5.0;
}
System.out.println(finalPrice);
```

What is printed as a result of executing the code segment?

(A)
20.0

(B)
25.0

(C)
70.0

(D)
75.0

153. Consider the following class declaration

```
public class SomeClass
{
  private int num;

  public SomeClass(int n)
  {
    num = n;
  }

  public void increment(int more)
  {
    num = num + more;
  }

  public int getNum()
  {
    return num;
  }
}
```

The following code segment appears in another class.

```
SomeClass one = new SomeClass(100);
SomeClass two = new SomeClass(100);
SomeClass three = one;

one.increment(200);

System.out.println(one.getNum() + " " + two.getNum() + " " +
                   three.getNum());
```

What is printed as a result of executing the code segment?

(A) `100 100 100`
(B) `300 100 100`
(C) `300 100 300`
(D) `300 300 300`

154. In the following code segment, str1 and str2 are String objects. The code segment is intended to print true if str1 and str2 are non-null strings that contain the same sequence of characters. The code segment is intended to print false otherwise.

```
boolean result = false;          // line 1
if(str1 != null && str2 != null)  // line 2
{
  result = str1 == str2;         // line 4
}
System.out.println(result);
```

Which of the following best explains the error, if any, in the code segment?

(A) In line 1, result should be initialized to true.
(B) In line 2, str1 and str2 should be compared to null with the equals method.
(C) In line 4, str1 and str2 should be compared to each other with the equals method.
(D) The code segment contains no errors and works as intended.

155. Consider the following code segment.

```
String str = "lookout";
String target = "o";
int j = str.indexOf(target);
while (j >= 0)
{
  str = str.substring(j);
  System.out.print(str + " ");

  str = str.substring(1);
```

```
  j = str.indexOf(target);
}
```

What is printed as a result of executing this code segment?

(A)
ookout okout out

(B)
ookout out

(C)
okout out

(D)
kout ut

156. Consider the following code segment.

```
String str = "abcdef";
for (int rep = 0; rep < str.length() - 1; rep++)
{
  System.out.print(str.substring(rep, rep + 2));
}
```

What is printed as a result of executing this code segment?

(A) `aabbccddeeff`
(B) `abbccddeef`
(C) `abcbcdcdedef`
(D) Nothing is printed because an `IndexOutOfBoundsException` is thrown.

157. Consider the following method.

```
/** Precondition: Strings one and two have the same length. */
public static String combine(String one, String two)
{
String res = "";
for (int k = 0; k < one.length(); k++)
{
    if (one.substring(k, k + 1).equals(two.substring(k, k + 1)))
    {
      res += one.substring(k, k + 1);
    }
```

```
    else
    {
      res += "0";
    }
  }
  return res;
  }
```

What is returned as a result of the call `combine("10110", "01100")` ?

(A) `"00000"`
(B) `"00100"`
(C) `"00101"`
(D) `"10110"`

158. Consider the following method, which is intended to count the number of times the letter `"A"` appears in the string `str`.

```
public static int countA(String str)
{
int count = 0;

while (str.length() > 0)
{
    int pos = str.indexOf("A");
    if (pos >= 0)
    {
      count++;
      /* missing code */
    }
    else
    {
      return count;
    }
}
return count;
}
```

Which of the following should be used to replace `/* missing code */` so that method `countA` will work as intended?

(A)
```
str = str.substring(0, pos + 1);
```

(B)
```
str = str.substring(pos - 1);
```

(C)
```
str = str.substring(pos);
```

(D)
```
str = str.substring(pos + 1);
```

159. Consider the following code segment.

```
String str = "abbcdddeff";
int j = 0;
int count = 0;
while (j < str.length() - 1)
{
  if (str.substring(j, j + 1).equals(str.substring(j + 1, j + 2)))
  {
    count++;
  }
  j++;
}
System.out.println(count);
```

What is printed as a result of executing this code segment?

(A)
2

(B)
3

(C)
4

(D)
6

160. Consider the following method.

```
public static int what(String str, String check)
{
int num = -1;
int len = check.length();
for (int k = 0; k + len <= str.length(); k++)
{
    String a = str.substring(k, k + len);
    if (a.equals(check))
    {
      num = k;
```

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```
        }
    }
    return num;
    }
```

Assume that `check` occurs at least once in `str`. Which of the following best describes the value returned by the `what` method?

(A) The number of times the string `check` occurs in `str`
(B) The index of the first occurrence of `check` inside `str`
(C) The index of the last occurrence of `check` inside `str`
(D) The number of substrings in `str` with the same length as `check`

161. Consider the following method.

```
String word = "Computer Science";
String str = "";

for (int k = 0; k < word.length(); k++)
{
  if (k % 3 == 0)
  {
    str = word.substring(k, k + 1) + str;
  }
}
System.out.println(str);
```

What is printed as a result of executing the code segment?

(A)
ci tm

(B)
eeStm

(C)
ncepC

(D)
eeSepC